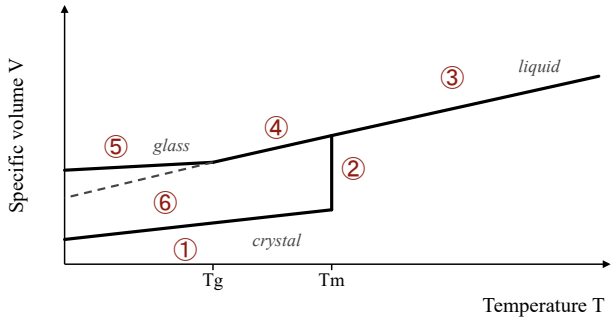
Advanced Materials — Exam Answer Sheet

42 past-exam questions · concise English answers in the wording of the lecture slides · ①②③ in the text match the numbers on each sketch (solid = draw this; dashed = reference).

1. Thermal transitions

Q21 *What's the definition of state function? Give the physical meaning of the second-order transition in terms of ΔV and ΔH?*

State function: depends only on the state, not on the path (H, S, U, A, G, P, V, T). W and Q depend on the path, so they are not state functions.



① Crystal: V increases slowly with T.

② Tm: V, H, S (first derivatives of G) **jump → 1st-order** transition (ΔV ≠ 0, ΔH ≠ 0).

③ Liquid: V increases with a larger slope.

④ Cool quickly: no crystallization, the liquid line continues below Tm.

⑤ Tg: V, H, S are **continuous (ΔV = 0, ΔH = 0)**; only the slopes change — α, Cp, κ (second derivatives of G) step → **2nd-order** transition.

⑥ (dashed) the liquid line extended, for comparison.

Slides: Polym Solid s046, s048, s083, s090

Q9 *How to determine the Tg of an amorphous polymer (give at least 2 ways)? Why a peak appears at the Tg point as the DMA spectrum was shown?*

① Volume (V–T): the slope of V vs T changes at Tg.

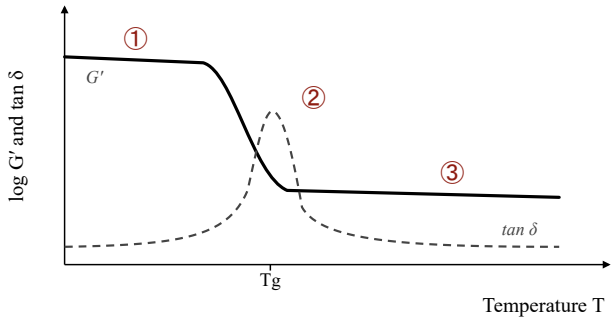
② DSC: a ΔCp step in the specific heat at Tg.

③ DMA: G′ drops from the glassy to the rubbery value; tan δ shows a maximum.

Why a DMA peak: in the glass transition region the segments start cooperative motions but these are sluggish. tan δ = G″/G′ shows a maximum when the storage modulus changes from a value characteristic of the glassy state to one characteristic of rubbers (sketch in Q37).

Slides: Polym Solid s083, s124; Mechanical Properties 1 s045–046

Q37 *Why a peak appears at the Tg point as the DMA spectrum was shown?*



① Below Tg (glassy): conformational changes are inhibited → G′ high, tan δ small.

② Glass transition region: cooperative segmental motions start but are sluggish → G′ drops from the glassy to the rubbery value → **tan δ = G″/G′ shows a maximum**.

③ Above Tg (rubbery plateau): G′ low and flat → tan δ small again.

Slides: Polym Solid s107 (PS tan δ peak ≈ 370 K), s124; Mechanical Properties 1 s045–046

Q19 *The melting temperature of PVDF is 174 °C, please estimate its T_g value.*

Tg ≈ 224 K (≈ −49 °C).

Crystalline polymers: Tg ≈ ½ to ⅔ Tm (in K). Tm = 174 + 273 = 447 K → Tg ≈ 224–298 K.
Taking ½: Tg ≈ 447 × ½ ≈ 224 K (−49 °C), close to the tabulated PVDF value −45 °C.

Slides: Polym Solid s080 (Table 4-3: PVDF Tg −45 °C)

Q8 *Please use Simha-Boyer equation to estimate the thermal expansion coefficient α_g of PS below Tg (Tg = 373 K, $\alpha_l = 5.5\times10^{-4}\ K^{-1}$)*

αg ≈ 2.47×10⁻⁴ K⁻¹.

Simha–Boyer: (αl − αg)Tg = 0.113
αg = αl − 0.113/Tg = 5.5×10⁻⁴ − 0.113/373 = 5.5×10⁻⁴ − 3.03×10⁻⁴ = 2.47×10⁻⁴ K⁻¹

Slides: Polym Solid s121 (Fig. 12.5, Simha–Boyer 0.113)

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2. Structure and Tg

Q4 *Which one of the following polymers has the highest T_g (1) PS (2) Poly(vinyl biphenyl) (3) Poly(vinyl naphthalene), explain.*

(2) Poly(vinyl biphenyl), Tg ≈ 145 °C.

Bulky pendant groups raise Tg through steric hindrance to bond rotations: PS 100 °C < poly(vinyl naphthalene) 135 °C < poly(vinyl biphenyl) 145 °C.

Slides: Polym Solid s098 (Bulky substituents)

Q5 *Which one of the following polymers has the highest T_g (1) poly(methyl acrylate) (2) poly(methyl methacrylate) (3) poly(pentyl acrylate)*

(2) PMMA (Tg = 378 K).

PMA 279 K → PMMA 378 K: the extra α-CH_3 on the backbone hinders bond rotation.
Flexible side chains increase the free volume through their effect on the packing of the chains, so Tg is lowered: methyl 279 > ethyl 249 > propyl 225 > butyl 218 K → poly(pentyl acrylate) is even lower.

Slides: Polym Solid s098, s099, s118–119 (Table 12.2)

Q6 *Which one of the following polymers has the highest T_g (1) polyethylene (2) polypropylene (3) polybutylene (4) polyhexylene (5) poly(4-methyl pentene)*

(5) Poly(4-methyl pentene), Tg = 302 K.

Flexible n-alkyl side chains: longer → more free volume → Tg lowered: PP 253 > polybutene 249 > polyhexene 223 K (PE 188 K).
The branched, bulky side group of poly(4-methyl pentene) causes steric hindrance to bond rotations → highest Tg.

Slides: Polym Solid s098, s099, s118 (Table 12.2)

Q7 *Which one of the following polymers has the lowest T_g (1) PMMA (2) PEMA (3) PPMA (4) PBMA*

(4) PBMA (Tg ≈ −2 °C).

Flexible substituents: PMMA 105 > PEMA 65 > PPMA 38 > PBMA −2 °C. These side chains increase the free volume through their effect on the packing of the chains, so Tg is lowered.

Slides: Polym Solid s099–100

Q25 *Which one of the following PU (with the same MW and diol content) should have the lowest Tg (1) HMI (2) H_{12}MDI (3) MDI (4) TDI*

(1) HMI.

HMI is $–(CH_2)_6–$ with no ring → flexible chain → lowest Tg. H_{12}MDI, MDI and TDI contain rings in the backbone → high energy barrier to rotations → chain stiffness → higher Tg.

Slides: Condensation polymerization s045 (Table 9-7); Polym Solid s096 (Chain stiffness)

Q10 *Estimate the Tg of 30 wt% di-n-butylphthalate (DBP) (Tg = 178 K) compounded with 70 wt% PVC (Tg = 355 K)?*

Tg ≈ 273 K (0 °C).

Fox equation: $1/Tg = w_1/Tg_1 + w_2/Tg_2 = 0.30/178 + 0.70/355 = 3.657\times10^{-3}\ K^{-1}$ → Tg ≈ 273 K. The diluent (plasticizer) lowers the Tg of PVC from 355 K.

Slides: Polym Solid s104 (Diluents), s105 (Fox equation)

Q11 *Which ones of the followings are miscible blends (multiple choices) (1) PC/PBT (2) PS/PPO (3) PVDF/PMA (4) PEO/PMMA*

(2), (3), (4).

PS/PPO, PVDF/PMA and PEO/PMMA are miscible blends: one phase → a single Tg. (1) PC/PBT is immiscible: well blended but still has two Tgs and a sharp melting point.

Slides: Polym Solid s106, s108, s112, s115

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3. Structure and Tm

Q16 *The PTFE has lower melt energy than the PE does. (2860 J/mole of PTFE vs. 3870 J/mole of PE) How come the Tm of PTFE is much higher than PE? (645 K vs. 410 K) (Hint: Table 11.1 can help you)*

PTFE has a much smaller ΔSf.

Tm = ΔHf/ΔSf → ΔSf = ΔHf/Tm: PE 3870/410 = 9.44; PTFE 2860/645 = 4.43 J mol^{-1} K^{-1} (about half).
PTFE has a larger trans–gauche energy difference and higher rotation barriers → stiff chain → small ΔSf → high Tm despite the lower ΔHf.

Slides: Polym Solid s064, s067–068; PE vs PTFE slides 15–16

Q12 *Why is the Tm of isotactic PP higher than that of syndiotactic PP (187 vs. 150)?*

Tm = ΔHf/ΔSf, and Tm rises with lamellar thickness: $Tm = Tm°[1 - 2\sigma e/(l\cdot\Delta hf)]$.

Both iPP and sPP are regular chains and crystallize. The isotactic helices pack into more stable crystals (larger ΔHf / thicker lamellae l) → higher Tm: 187 > 150 °C.

Slides: Polym Solid s015, s058, s064, s153

Q22 *Will the "regularity of configuration" affect more on Tg or Tm? Please use PP as an example to illustrate your point.*

Regularity of configuration affects Tm more.

Tm: crystallization requires a regular chain. Isotactic and syndiotactic PP crystallize (have Tm); atactic PP cannot crystallize.

Tg: set by chain stiffness and bulky pendant groups; all PP have the same $–CH_3$ (atactic PP Tg ≈ −10 °C).

Slides: Polym Solid s015, s098, s153

Q13 *Place the following polyurethane structures in order of decreasing Tm? (1) –HN–$(CH_2)_6$–NH–CO–O–$(CH_2)_3$–C_6H_4–$(CH_2)_3$–O–CO– (2) –HN–$(CH_2)_8$–NH–CO–O–$(CH_2)_2$–C_6H_4–$(CH_2)_2$–O–CO– (3) –HN–$(CH_2)_8$–NH–CO–O–$(CH_2)_{10}$–O–CO–*

(1) > (2) > (3), by Tm = ΔHf/ΔSf.

① (1), (2): phenylene ring → stiff → ΔSf small → Tm high.

② (3): no ring, many flexible CH_2 → ΔSf large; fewest urethane H-bonds → ΔHf small → lowest.

③ (1) vs (2): same ring and same CH_2 count; (1) has its urethane groups closer together → more concentrated H-bonding → larger ΔHf → (1) > (2).

Slides: Polym Solid s063 (H bonds), s067–069, s093 (Table 4-8)

Q14 *Place the following Tm of the polyolefins with different side chain structures in a decreasing order and give a brief explanation (1) –CH_3 (2) –CH_2–$C(CH_3)_2$–CH_2CH_3 (3) –$CH_2CH_2CH_3$ (4) –$CH_2CH_2CH_2CH_3$ (5) –CH_2–$CH(CH_3)$–CH_2CH_3*

(2) > (5) > (1) > (3) > (4), by Tm = ΔHf/ΔSf.

Bulky, branched side groups close to the backbone → stiff chain → ΔSf small → Tm high; (2) is more crowded than (5).
Longer flexible linear side chains → ΔSf large → Tm lower: $–CH_3$ > propyl > butyl.

Slides: Polym Solid s064, s067–068, s079 (Fig. 7-4)

Q15 *Please list the Tm of the followings with the same MW in a decreasing order: (a) –[Ph–CO–Ph–O]– (b) –[Ph–CO–Ph–CO–Ph–O]– (c) –[Ph–O–Ph–CO–Ph–O]– (d) –[Ph–Ph–O–Ph–CO–Ph–O]–*

(d) > (b) > (a) > (c) (689 > 657 > 640 > 628 K, Table 15.11).

Stiff → ΔSf small → Tm high. (d) has a directly linked biphenyl (stiffest); (b) has more C=O than (a); (c) has more flexible ether –O– links → lowest.

Slides: Polym Solid s067–068, s093, s163–164 (Table 15.11)

Q17 *Place the following Tm of nylon in a decreasing order and give brief explanation, Nylon 8, Nylon 9, Nylon 10, and Nylon 11.*

Nylon 9 > 8 > 11 > 10 (482 > 473 > 463 > 461 K, Table 15.3).

Larger n → fewer amide groups → less interchain hydrogen bonding → Tm decreases. Odd and even n allow different H-bond alignment, so the order is not strictly monotonic.

Slides: Polym Solid s019–020, s136 (Table 15.3)

Q18 *Place the following Tm of polyester in a decreasing order $[(CH_2)_mCOO]$; m = 2, 3, 4, 5, 6*

m = 2 > 3 = 5 > 4 > 6 (395 > 335 = 335 > 329 > 325 K, Table 15.1).

Larger m → lower ester-group density, more flexible CH_2 → Tm decreases (with a small odd/even fluctuation).

Slides: Polym Solid s135 (Table 15.1)

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4. Crystallinity and barrier properties

Q3 *How to determine the crystallinity of a polymer?*

① Density: φ = (ρ − ρa)/(ρc − ρa).

② X-ray diffraction: sharp crystalline peaks on the broad amorphous halo.

③ DSC: the crystalline-melting endotherm.

Slides: Polym Solid s004 (Table 4-5), s010, s050 (Fig. 4-14)

Q20 *How to determine the crystallinity of a polymer? How to determine the lamellae thickness of a polymeric crystal?*

Crystallinity: density $\varphi = (\rho - \rho a)/(\rho c - \rho a)$; X-ray diffraction; DSC melting endotherm.

Lamellar thickness: measure Tm (DSC) and use the Thompson–Gibbs rule:

$$Tm = Tm^\circ[1 - 2\sigma e/(l \cdot \Delta hf)] \rightarrow l = 2\sigma e Tm^\circ / [\Delta hf(Tm^\circ - Tm)]$$

Thicker lamellae melt closer to Tm°. Typical $l \approx 150$ Å.

Slides: Polym Solid s004, s010, s050, s054–058, s034–035

Q38 *Which of the following polymers would expect to have the best barrier properties (hint: crystalline materials perform better than non-crystalline materials) (1) Atactic polystyrene (2) A random ethylene/propylene copolymer (50/50) composition. (3) mCOC (4) HDPE*

(4) HDPE. Crystallization requires a regular chain; crystalline materials are better barriers (hint).

HDPE: linear, regular $–CH_2–$ chain → highest crystallinity (density 0.94–0.97 g/cm³).

Atactic PS: atactic → cannot crystallize.

Random E/P copolymer: random sequence → irregular → cannot crystallize.

mCOC: random bulky cyclic units → irregular → cannot crystallize.

Slides: Polym Solid s015, s150, s153, s156 (Table 15.7)

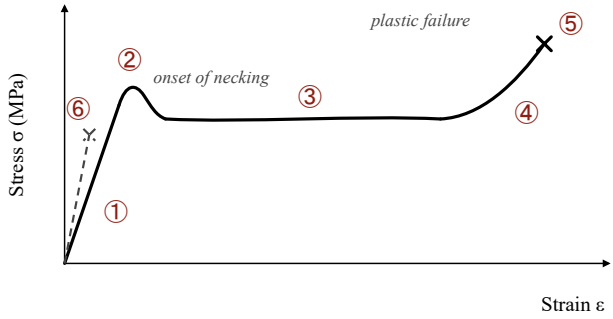
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5. Stress–strain curves

Q1 *Please give a whole picture of the stress-strain curves for a semi-crystalline polymer with a polymer-morphology point of view.*

Semi-crystalline polymer (plastic response):

① Initial linear rise: amorphous regions elongate.
② Yield point, onset of necking: crystalline regions align (tilt toward the load).
③ Plateau: the neck extends along the sample; crystalline regions slide (lamellae break into blocks).
④ Near failure: chains and blocks are aligned with the load → stress rises.
⑤ Plastic failure (×).
⑥ (dashed) brittle failure: no yield, no necking.

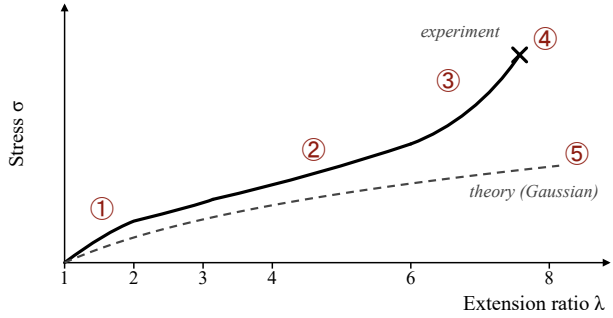


Slides: Mechanical Properties 1 s024–026; Polym Solid s042

Q2 *Please give a whole picture of the stress-strain curves for a rubbery polymer with a molecular conformation point of view.*

Stretching lowers the conformational entropy → restoring force; theory $f = NkT(\lambda - \lambda^{-2})$.

① Low λ: obeys Hooke's law and the theory; low modulus (coiled chains straighten).
② $\lambda \gtrsim 3$: above the theory (Mooney–Rivlin fits better).
③ High λ: steep rise — chains are straight, still cross-linked.
④ Fracture. Deformation is reversible; no yield peak.
⑤ (dashed) Gaussian theory.



Slides: Elasticity s038, s047, s053, s063

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6. Rubber elasticity

Q39 *In deriving the rubber elastic theory, what assumptions were used.*

① Gaussian chains (large N).

② Only entropy: the small contribution of internal energy is ignored.

③ Affine deformation: chains deform in exact proportion to the sample.

④ No change in volume: $\lambda_1\lambda_2\lambda_3 = 1$.

⑤ Model network: same crosslink functionality, no dangling ends.

⑥ Constant temperature (reversible).

Result: $f = NkT(\lambda - \lambda^{-2})$, $E = NkT$.

Slides: Elasticity s029–037, s053, s063

Q33 *Pick up the right choices as follows (multiple choices) (1) Endothermic under tension (2) Modulus increases with temperature (3) Elasticity comes mainly from the entropy change (4) Adiabatic process is suitable to describe rubber under a large deformation?*

(2), (3), (4).

(1) Wrong: stretching reduces conformational entropy → heat is given off, not absorbed.

(2) Right: T↑ → rubber modulus Gr↑ (Hookean solid: Gh↓).

(3) Right: $f \approx -T(dS/dl)$ for elastomer networks.

(4) Right: "Rubber: reversible adiabatic extension."

Slides: Elasticity s028, s063, s075–079

Q34 *Pick up the right choices as follows (multiple choices) for $f = (\partial U/\partial l)_T - T(\partial S/\partial l)_T$ (1) $(\partial U/\partial l)_T$ is valid for crystalline and glassy state (2) $(\partial U/\partial l)_T$ represents the Hook's elasticity (3) $T(\partial S/\partial l)_T$ represents the Rubber elasticity (4) $(\partial U/\partial l)_T$ increases with temperature (5) For a given elongation, the $(\partial U/\partial l)_T$ is 10^4 higher than the $T(\partial S/\partial l)_T$.*

(1), (2), (3).

(1)(2) Right: crystalline & glassy solids $f \approx (dU/dl)$ → energy (Hookean) elasticity.

(3) Right: elastomer networks $f \approx -T(dS/dl)$ → rubber elasticity.

(4) Wrong: fU stays nearly constant with T; $fS = T(df/dT)$ increases.

(5) Wrong: in rubber $fU/f \approx 0.10$–0.17 (Table 9.2) → the energy term is the smaller one.

Slides: Elasticity s028, s059, s072–078

Q42 *Which one of the following statement regarding polyisoprene (natural rubber) is right (1) Natural rubber is mainly a trans form (2) The trans form has lower Tg (3) the repeat distance of cis is 0.88 nm (4) The Golfball covers can be made by Polyisoprene in trans form*

(4).

(1) Wrong: natural rubber is cis-1,4-polyisoprene.

(2) Wrong: trans has the higher Tg (220 vs 200 K).

(3) Wrong: cis repeat distance is 0.81 nm; 0.88 nm is trans.

(4) Right: trans is the harder, more crystallizable form (higher Tg, Tm 347 vs 301 K, larger ΔHu) → suitable for hard golf-ball covers.

Slides: Elasticity s005–007; Bond and Conformation s030 (Table 11.1)

Q35 *Please give your brief reasons why the space shuttle Challenge exploded in 1988.*

The fluoroelastomer O-ring seals the joint by its **resiliency** (return to original size and shape after deformation). **When cold it was slow to return to its original shape** (Feynman, iced water) → the joint was not sealed → hot gas leaked and caused **erosion** → the leaking flame ignited the hydrogen fuel in the external tank. (Accident: 28 Jan 1986.)

Slides: Mechanical Properties 1 s002–003

Q36 *What is the "manufacturing secret" to make the "thermoplastic elastomer"?*

Tri-block copolymer: one soft elastomeric segment + two hard (glassy or crystalline) blocks, e.g. polystyrene-block-polybutadiene-block-polystyrene.

① Processing: both segments above Tg → the material flows.

② On cooling: separation of the phases into two domain types creates **physical crosslinks**.

③ The soft segment between the domains gives rubber elasticity — no chemical crosslinking needed.

Slides: Elasticity s064, s066; Polym Solid s141 (Fig. 15.6)

Q24 *Why does a SBS triblock copolymer exhibit a non-Newtonian and even highly elastic behavior at low temperature (< 423 K) but show a Newtonian behavior at 448 K?*

Below 423 K: the glassy PS blocks separate into domains → **physical crosslinks** → network → non-Newtonian, highly elastic.

At 448 K: the domains no longer hold the chains (both segments flow) → no network → Newtonian melt.

Slides: Elasticity s066; Polym Solid s141 (Fig. 15.6)

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7. Viscoelasticity

Q27 *List four elastic characters of a viscoelastic polymer.*

① Normal stress effect (rod climbing): the polymer climbs the rod; a Newtonian fluid dips.

② Die swell: the extrudate is larger than the die (fading memory).

③ Tubeless siphon: the polymer keeps flowing after the tube leaves the liquid.

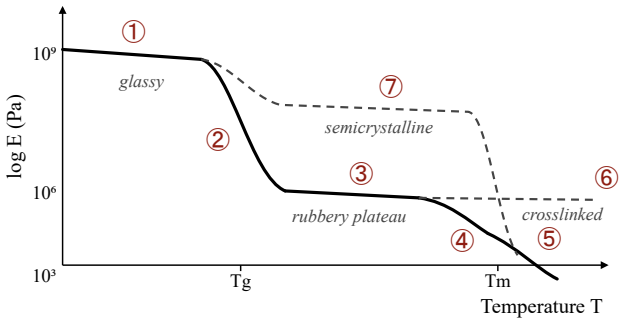
④ Secondary flow: vortices in the corners of a contraction flow.

(Also: elastic recoil.) Chains disentangle and stretch out → elastic component of polymer melts.

Slides: Rheo. Intro s006, s008, s010, s012, s014; Viscoelasticity s012

Q31 *Pick up the right choices for describing the Relaxation Modulus vs. Temp. (Master curve) as follows (multiple choices) (1) Glassy modulus is 10^2 to 10^3 times higher than the Rubbery Modulus (2) Rubbery Modulus increases with the crosslinking degree (3) Rubbery Modulus increases with the crystallinity (4) the modulus of semicrystalline polymer remains independent of temperature if the crystallinity remains unchanged. (5) The crystallinity has little effect on the Glassy modulus*

All five, (1)–(5).



(1) Glassy (①) vs rubbery plateau (③): about 10^3 times (Fig. 15.9).

(2) Crosslinks keep the plateau (⑥, no terminal flow); E = NkT → more crosslinked chains N → higher modulus.

(3) Crystallinity raises the modulus between Tg and Tm (⑦).

(4) "Directly proportional to the degree of crystallinity and independent of temperature if the crystalline order remains unchanged."

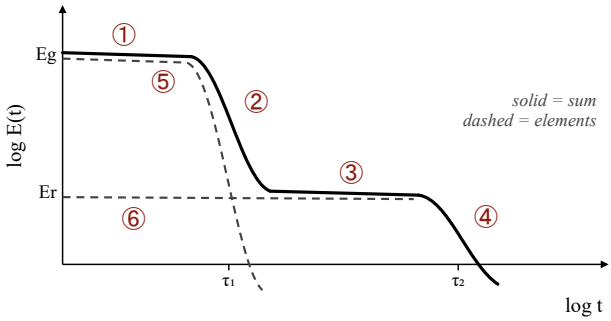
(5) "Crystallinity only affects the response from Tg to Tm; below Tg the effect on the modulus is small."

① glassy ② glass transition ③ rubbery plateau ④ rubbery flow ⑤ terminal flow ⑥ cross-linked ⑦ crystalline

Slides: Polym Solid s143 (Fig. 15.9), s145; Viscoelasticity s022–026; Elasticity s037

Q32 *How many Maxwell models in parallel at least can you apply to describe a typical Master curve (Relaxation Modulus vs. Temperature)? How to match the curve?*

Two Maxwell elements in parallel (Maxwell–Wiechert, n = 2). A single Maxwell model has a single relaxation time → only one drop.



$$E(t) = E_1 \exp(-t/\tau_1) + E_2 \exp(-t/\tau_2), \quad \tau = \eta/E, \quad \tau_2 \gg \tau_1$$

① Short t: glassy modulus Eg ≈ $E_1 + E_2$.

② At τ_1: element 1 relaxes → first drop.

③ Rubbery plateau Er = E_2.

④ At τ_2: element 2 relaxes → terminal flow.

⑤⑥ (dashed) the two single elements.

Matching: step heights → E_1, E_2; drop positions → τ_1, τ_2.

Slides: Viscoelastic Modelling s010–013, s021, s024–026

Q40 *The viscosity of polystyrene is 10^{12} Pa·s at 373 K. Determine the viscosity when the temperature increases to 473 K (WLF eq., $a_1 = 17.44$, $a_2 = 51.6$)*

η(473 K) ≈ 3.1 Pa·s.

$\eta(T) = a_T\,\eta(Tg)$; $\log a_T = -17.44(T - Tg)/[51.6 + (T - Tg)]$, $T - Tg = 100$ K

$\log a_T = -17.44 \times 100/151.6 = -11.5 \rightarrow a_T = 3.1\times10^{-12}$

$\eta = 3.1\times10^{-12} \times 10^{12} \approx 3.1$ Pa·s

Slides: Polym Solid s129, s132–133

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8. Melt rheology

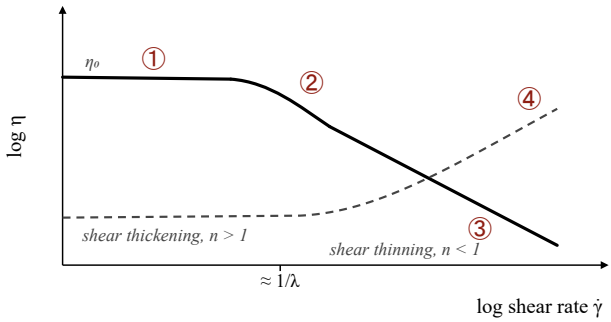
Q26 *What is shear thinning? What is shear thickening? Why a polymer behaves a shear thinning rather than shear thickening.*

Power-law fluid: $\eta = m \cdot \dot{\gamma}^{n-1}$; n = 1 Newtonian.

Shear thinning (pseudoplastic): n < 1, η decreases as $\dot{\gamma}$ increases (③).

Shear thickening (dilatant): n > 1, η increases as $\dot{\gamma}$ increases (④, dashed).

Polymer → thinning: under shear the random coil is stretched and the chains disentangle and align with the flow → η decreases.



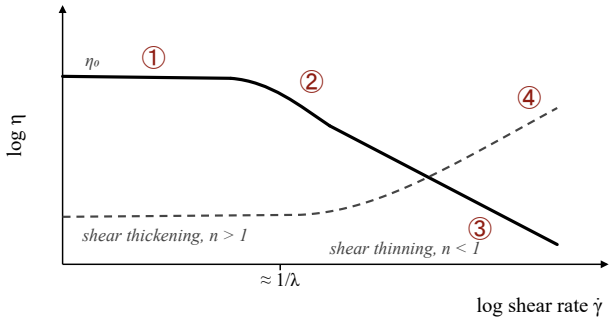
Slides: Newtonian Fluid s017, s019–021; Viscoelasticity s002, s012

Q23 *Why does a polymer melt usually exhibit a shear thinning behavior?*

① Low $\dot{\gamma}$ (De ≪ 1): chains relax back to random coils → zero shear rate viscosity η_0 (plateau).

② $\dot{\gamma} \approx 1/\lambda$: the curve turns down.

③ High $\dot{\gamma}$: chains cannot relax → stretched, disentangled, aligned with the flow → η decreases (shear thinning); processing shear rates lie here.



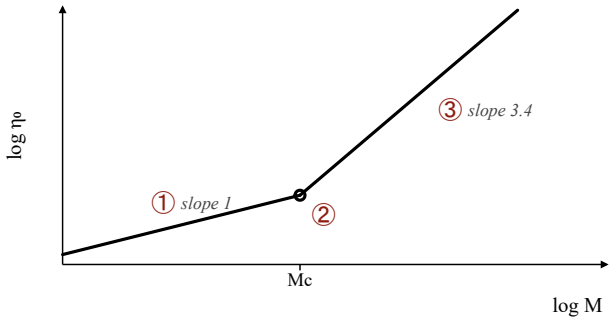
Slides: Viscoelasticity s002–005, s008–009, s012; Newtonian Fluid s019

Q28 *What is the relationship between the molecular weight and the melt viscosity?*

① M < Mc: $\eta_0 \propto M$ (slope 1) — viscosity simply follows chain length.

② At the critical molecular weight Mc (= 2–3 Me) chains start to become tangled up.

③ M > Mc: entanglements, reptation → $\eta_0 \propto M^{3.4}$ (slope 3.4).



Slides: Fluid 2 s072, s074–076; Viscoelasticity s053

Q29 *What will the melt viscosity be affected by (1) temperature (2) pressure (3) shear rate (4) molecular weight (5) plasticizer wt%*

All five. Free volume controls viscosity: Doolittle $\eta = A \exp(B/Vf)$.

(1) T↑ → Vf↑ → η↓ (WLF).

(2) Pressure↑ → Vf↓ → η↑.

(3) $\dot{\gamma}$↑ beyond the zero-shear plateau → η↓ (shear thinning).

(4) M↑ → η↑: $\eta_0 \propto M$ (< Mc), $\propto M^{3.4}$ (> Mc).

(5) Plasticizer↑ → Vf↑ → η↓.

Slides: Polym Solid s088, s129; Viscoelasticity s009; Fluid 2 s072–076

Q30 *Please explain the Cox-Merz rule.*

Cox–Merz rule: the magnitude of the complex viscosity at angular frequency ω equals the steady-shear viscosity at the same numerical shear rate:

$$\eta(\dot{\gamma}) = |\eta^*(\omega)| \text{ at } \dot{\gamma} = \omega; \quad |\eta^*| = \sqrt{(\eta'^2 + \eta''^2)}, \eta' = G''/\omega, \eta'' = G'/\omega$$

So frequency sweeps (G′, G″) give the flow curve; with a master curve (TTS) the range is extended (PS example: |η*| and η give the same values).

Slides: Viscoelasticity s046–047, s050, s054–055

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9. Fracture mechanics

Q41 *Considering an aluminum plate has Kc of 44 MPa $m^{1/2}$, thickness 20 mm and width 100 mm, with 40 mm crack. What is the critical limit for the plate of load (kN)? (the geometry factor Y = 2.1)*

Fc ≈ 118 kN.

Failure when K = Kc; K = (YF/(WB))√(πa):
Fc = Kc·W·B / (Y√(πa)) = (44×10⁶ × 0.100 × 0.020) / (2.1 × √(π × 0.040))
= 88 000 / 0.744 ≈ 1.18×10⁵ N ≈ 118 kN

Slides: Brittle & Ductile s032, s034–035
